

Individual Reflective Document

Introduction

This document analyses my experiences in the module Secure Software Development.

Reflections will be made using Jasper's Evaluation, Reflection and Action (ERA) Cycle (Jasper, 2013). I have modified the cycle to begin with an Action phase, by creating a Personal Development Plan (PDP) based on previous reflections.

Consequently, I will reflect at micro level on module tasks and at macro level on the PDP. Through this I hope to achieve double-loop reflective learning, considered an effective way to evaluate biases and motivations (Gibbin et al., 2016).

PDP Goals

In previous modules I was disappointed in my engagement level with the cohort, all more experienced in the computing field than myself. Therefore, the primary goal of this PDP was to actively participate in seminars and take opportunities to involve myself with group tasks. As an introvert, this prospect made me slightly uncomfortable, but this can be considered positive for development (Wooley and Fishbach, 2022).

A secondary goal was to improve the technical skills as I struggled to complete technical tasks on time in past assignments. I hoped to add a more professional style to any coding work. I also felt that in previous modules I gained a lot of theoretical knowledge but did little to improve on a practical level.

Development Team Project

We were tasked with designing a secure music storage application in a group. The project started positively; the first meeting was productive; tasks were created and assigned main and support contributors. Following my PDP I took an active role and put myself forward as a contributor for multiple tasks. Most of the group worked collaboratively, uploading work to a shared folder for all group members to make suggestions on. We ended with a clear plan for the application.

The project was successful, the learning outcome of identifying and managing security tasks was met, and I was pleased with my input. However, the team dynamic was frustrating; some team members took advantage of my willingness to contribute, and the overall distribution of work was uneven.

In future, I would take time to assess individuals in the team and fit them into Belbin's team building model (Belbin and Brown, 2022) to create a more effective working arrangement.

Individual Development Project

As individuals we implemented the team design in Python; the learning outcome was to meet the brief and critically evaluate solutions. The task fit my PDP goal of utilising theoretical knowledge practically.

This began with studying coding practices, then proceeded with programming. Static code analysis was performed with tools such as Pylint and Bandit to ensure quality code.

I was apprehensive about the task's scale, but the process was rewarding. My understanding of coding principles and style improved, and I was introduced to testing with linters. The overall task gave me the feeling that I would enjoy working in IT professionally and I was proud of the result.

Despite this, I did struggle to complete the task on time, and some aspects felt unfinished. For future projects I would create a series of deadlines for the project to keep myself accountable and track my progress.

Conclusion

I did achieve the goals of my PDP, and I was generally pleased with my progress through the module. However, I couldn't participate fully in all the module's formative tasks due to the time pressure caused by taking on too much in the group project.

Furthermore, I believe that my PDP was only reactive to prior events and that there should be more balance between response to difficulties and proactive development – positively identifying a skill that I would like to obtain. Accordingly, I should have included a goal that was more specific to software development.

For the next module, I would have one goal, improved time management, linked to the results of this reflection, and add a specific goal in the first two weeks of the new module.

References

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